

# RONALDO MENEZES

---

University of Exeter • Department of Computer Science • Exeter, UK  
r.menezes@exeter.ac.uk • ronaldomenezes@gmail.com • BlueSky:  
@ronaldomenezes.bsky.social

## EDUCATION

---

### University of York, England, UK

DPhil in Computer Science, 2000

Thesis: *Resource Management in Open Distributed Tuple Space Systems*

Advisor: Alan Wood

### State University of Campinas, Brazil

MSc in Computer Science (Thesis Option), 1995

Thesis: *A Study on Models of Parallel Computation*

Advisor: João Carlos Setubal

### University of Fortaleza, Brazil

BSc in Computer Science, 1992

## Scholarships

- PhD Scholarship (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Brazil) August 1995 – August 1999
- MSc Scholarship (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Brazil) January 1993 – December 1994

## RESEARCH INTERESTS

---

Network Science, Complex Systems, Human Mobility, Urban Systems, Data Science, Social Network Analysis, Self-Organisation, Swarm Intelligence, Biologically-Inspired Computing, Science of Science, Inequality Modelling.

## SELECTED RESEARCH GRANTS

---

### ◇ *Rapid Development of Novel Optimisation Tools for Transport and Energy Networks*

PI: Ed Keedwell; Co-PI: Ronaldo Menezes

Funded by Innovate UK, 2021–2024, £166,580

### ◇ *Methods for the Identification of Robbery, Theft of Artworks*

PI: Douglas Ferreira; External Collaborator: Ronaldo Menezes

Funded by FAPERJ-Brazil, 2019–2022, R\$250,000 (£35,000 approx.)

### ◇ *Understanding Human Movement in the Healthcare Sector using AI*

PI: Jacq Christmas; Co-PI: Ronaldo Menezes

Funded by Innovate UK, 2020–2022, £166,580

### ◇ *Hot Spot Policing Randomized Control Trial in the City of Exeter*

PI: Ronaldo Menezes; Co-PIs: Marcos Oliveira, Jo Arundale

Funded by Policing Lab, 2020–2021, £29,728

- ◇ *Identification of Human Mobility Modes using Socio-Spatio-Temporal Predictive Models*  
 PI: Gourab Ghoshal; Co-PIs: Ronaldo Menezes, Brooke Welles  
 Funded by US Army Research Office, 2018–2021, US\$500,000
- ◇ *Establishment of IANE (Northeastern Center for Artificial Intelligence), part of the IARA project*  
 PI: Fernando Buarque; External Collaborator: Ronaldo Menezes  
 Funded by a Private Consortium of Industries and Government Agencies, 2020, R\$7,000,000 (£1,000,000 approx.)
- ◇ *Automated Model Build for Decarbonisation and Climate Resilience*  
 PI: Ronaldo Menezes; Co-PIs: Federico Botta, Marcos Oliveira  
 Funded by Innovate UK, 2021, £90,751
- ◇ *Development of App and Web Interface for Automated Anuran Recognition and Mapping*  
 PI: Mark Bush; Co-PIs: Ronaldo Menezes, Eraldo Ribeiro  
 Funded by NSF, 2012–2018, \$348,000
- ◇ *Understanding the Predictability Limits in Human Mobility and Social Tie Formation in the Absence of Complete Data*  
 PIs: Gourab Ghoshal, Ronaldo Menezes  
 Funded by Army Research Office, 2016, \$60,000
- ◇ *InToPerformance: Influence of Network Topology to the Performance of Wireless Sensor Networks*  
 PI: Angelo Brayner and Ronaldo Menezes  
 Funded by Funcap (Brazil), 2013–2014, \$19,384
- ◇ *Generating Networks from Government Websites*  
 PI: Ronaldo Menezes  
 Funded by Harvard University, 2013, \$13,834
- ◇ *Exploring the Use of Nature-Inspired Algorithms in Software Testing Tools*  
 PI: Ronaldo Menezes  
 Funded by CNPq (Brazil), 2005–2007, \$110,000
- ◇ *Proactive, Selective Behavior Filtering with Data Restoration: A New Technique for Runtime Neutralization of Malicious Mobile Code*  
 PI: James Whittaker; Co-PIs: Ronaldo Menezes, Cem Kaner, Michael Thomason, Jens Gregor and Gerald Marin  
 Funded by Office of Naval Research, 2001–2005, \$4,000,000

## PUBLICATIONS

---

### Books

- ◇ *Methods of Statistical Physics Applied to Seismology: A Study of Earthquakes from the Viewpoint of Complex Networks*  
 D. Ferreira, A. Papa, R. Menezes. LAP LAMBERT Academic Publishing, 2016.

### Books Edited

- ◇ *Complex Networks XVI*  
 M. Macedo, A. Cardillo, W. Franco, A. Brayner, R. Menezes. Springer Nature. 2025
- ◇ *Complex Networks XV*  
 F. Botta, M. Macedo, H. Barbosa, R. Menezes. Springer Nature. 2024
- ◇ *Complex Networks XIV*

- A. Teixeira, F. Botta, J. F. F. Mendes, R. Menezes, G. Mangioni. Springer Nature. 2023
- ◇ *Complex Networks XIII*  
D. Pacheco, A. Teixeira, H. Barbosa, R. Menezes, G. Mangioni. Springer Nature. 2022
  - ◇ *Complex Networks XII*  
A. Teixeira, D. Pacheco, M. Oliveira, H. Barbosa, B. Gonçalves, R. Menezes. Springer Nature. 2021
  - ◇ *Complex Networks XI*  
H. Barbosa, J. Gomez-Gardenes, B. Gonçalves, G. Mangioni, R. Menezes, M. Oliveira. Springer Nature. 2020
  - ◇ *Proceedings of the Intelligent Data Engineering and Automated Learning–IDEAL 2019: 20th International Conference*  
H. Yin, D. Camacho, P. Tino, A. J. Tallón-Ballesteros, R. Menezes, R. Allmendinger. Manchester, UK, November 14–16, 2019, Springer Nature.
  - ◇ *Complex Networks VIII*  
B. Gonçalves, R. Menezes, R. Sinatra, V. Zlatic. Springer International Publishing. 2017
  - ◇ *Complex Networks VII*  
H. Cherifi, B. Gonçalves, R. Menezes, R. Sinatra. Studies in Computational Intelligence, Vol. 644. 2016
  - ◇ *Complex Networks V*  
P. Contucci, R. Menezes, A. Omicini, J. Poncela-Casasnovas. Studies in Computational Intelligence, Vol. 549. 2014
  - ◇ *Complex Networks IV*  
R. Menezes, A. Evsukoff and M. C. González. Studies in Computational Intelligence, Vol. 424. 2013
  - ◇ *Complex Networks: Second International Workshop, CompleNet 2010*  
L. F. Costa, A. Evsukoff, G. Mangioni and R. Menezes. Communications in Computer and Information Science, Vol. 116. 2011
  - ◇ *Complex Networks: Results of the 1st International Workshop on Complex Networks (CompleNet 2009)*  
R. Menezes, S. Fortunato, G. Mangioni and V. Nicosia. Studies in Computational Intelligence, Vol. 207. 2009

## BOOK CHAPTERS

- ◇ *Towards intelligent dynamics of an active transport system for biking*  
A. Medaglia, M. Wilches-Mogollon, O. Sarmiento, F. Montes, L. Guzman, M. Sanchez-Silva, R. Menezes, D. Hidalgo, K. Parra, A. Useche, H. Ochoa-Montero, N. Rodríguez, L. Salamanca. In *Safer Complex Systems Case Studies*. Royal Academy of Engineering. May 2022.
- ◇ *Networks in Rural Sociology*  
Ronaldo Menezes and Pramod Divakarmurthy. In *Encyclopedia on Social Network Analysis and Mining*, eds. Reda Alhajj and Jon Rokne. August 2014.
- ◇ *Simulating Crime Against Properties Using Swarm Intelligence and Social Networks*  
V. Furtado, A. Melo, A. Coelho, R. Menezes and M. Belchior. *Artificial Crime Analysis Systems*. IGI Global, 2008. 300–318.
- ◇ *Scalability in Linda-like Coordination Systems*  
R. Menezes, R. Tolksdorf and A. Wood. In *Coordination of Internet Agents: Models, Technologies and Applications*, eds. A. Omicini, F. Zambonelli, M. Klush and R. Tolksdorf. Springer-Verlag. 2001

## UNDER REVIEW AND ONGOING

- ◇ *Systematic comparison of gender inequality in scientific rankings across disciplines*  
A. M. Jaramillo, M. Macedo, M. Oliveira, F. Karimi, R. Menezes. arXiv preprint arXiv:2501.13061 (under review *Nature Human Behaviour*)
- ◇ *The parenthood effect in urban mobility*  
M. Macedo, R. Menezes, A. Cardillo. arXiv preprint arXiv:2501.02299 (under review *Nature Human Behaviour*)
- ◇ *Super-Linear Growth and Rising Inequality in Online Social Communities: Insights from Reddit*  
G. Machado, D. Pacheco, R. Menezes, G. Baxter. arXiv preprint arXiv:2503.02661 (under review *EPJ Data Science*)

## JOURNAL

- ◇ *Validating urban scaling laws through mobile phone data: A continental-scale analysis of Brazil's largest cities*  
R. S. Alencar, F. L. Ribeiro, H. Samaniego, R. Menezes, A. G. Evsukoff. *Computational Urban Science* 6 (1), 21. 2026
- ◇ *Natural language processing of biomedical text to map and prioritize protein–disease associations in HFP EF*  
C. Santana, C. Mukherjee, A. Quazi, R. Menezes, V. Filkov, D. Sigdel, H. Choi, et al. *Computers in Biology and Medicine* 205, 111599. 2026
- ◇ *Vector fields as a framework for modelling the mobility of commodities*  
S. Farokhnejad, A. S. da Mata, M. Macedo, R. Menezes. *PLoS One* 21 (3), e0340109. 2026
- ◇ *Not all roads are created equal: network science shows some highways matter more for Brazil's connectivity*  
J. Taveira, R. Menezes, F. Buarque. *Applied Network Science*. 2026
- ◇ *Geographical isolation as a driver of political violence in African cities*  
R. Prieto-Curiel, R. Menezes. *Nature Communications* 16, Article 10049. 2025
- ◇ *Beyond proximity: A review and framework to further understanding of greenspace accessibility in the x-minute city*  
A. Schendl, J. M. Bullock, R. Menezes, S. Willcock. *People and Nature*. 2025
- ◇ *A data-driven supervised machine learning approach to estimating global ambient air pollution concentrations with associated prediction intervals*  
L. J. Berrisford, H. Barbosa, R. Menezes. *Royal Society Open Science* 12 (7), 241288. 2025
- ◇ *The impact of the social exposome in cardiovascular health and disease*  
K. El Mekkaoui, C. Mukherjee, C. Santana, H. Silva, A. Quazi, R. Menezes. *The Journal of Precision Medicine: Health and Disease*. 2025
- ◇ *A framework for scalable ambient air pollution concentration estimation*  
L. J. Berrisford, L. S. Neal, H. J. Buttery, B. R. Evans, R. Menezes. *Environmental Data Science*, Volume 4, e17. 2025
- ◇ *Mobility and transit segregation in urban spaces*  
N. Iyer, R. Menezes, H. Barbosa. *Environment and Planning B: Urban Analytics and City Science* 51 (7), 1496–1512. 2024
- ◇ *The role of transport systems in housing insecurity: a mobility-based analysis*  
N. Iyer, R. Menezes, H. Barbosa. *EPJ Data Science* 13 (1), 49. 2024
- ◇ *Dynamic predictability and activity-location contexts in human mobility*  
B. Poudyal, D. Pacheco, M. Oliveira, Z. Chen, H. S. Barbosa, R. Menezes, G. Ghoshal. *Royal*

- ◇ *Dynamics of consumption of child sexual abuse material on the dark web*  
P. Divakarmurthy, B. Cunha, J. Passoud, R. Menezes, M. Oliveira. PLoS One 19 (7), e0306516. 2024
- ◇ *Environmental Insights: Democratizing access to ambient air pollution data and predictive analytics with an open-source Python package*  
L. J. Berrisford, R. Menezes. Environmental Modelling & Software, Volume 179, 106131. 2024
- ◇ *Estimating annual ambient air pollution using structural properties of road networks*  
L. J. Berrisford, E. Ribeiro, R. Menezes. Environment and Planning B: Urban Analytics and City Science 51 (9), 2031–2054. 2024
- ◇ *The structure of segregation in co-authorship networks and its impact on scientific production*  
A. M. Jaramillo, H. T. P. Williams, N. Perra, R. Menezes. EPJ Data Science 12 (1), 47. 2023
- ◇ *Transport and Mobility Segregation in Urban Spaces*  
N. Iyer, R. Menezes, H. Barbosa. International Journal of Population Data Science 8 (3). 2023
- ◇ *COVID-19 is linked to changes in the time–space dimension of human mobility*  
C. Santana, F. Botta, H. Barbosa, F. Privitera, R. Menezes, R. Di Clemente. Nature Human Behaviour, 1–11. 2023
- ◇ *Contrasting social and non-social sources of predictability in human mobility*  
Z. Chen, S. Kelty, A. Evsukoff, B. F. Welles, J. P. Bagrow, R. Menezes, G. Ghoshal. Nature Communications, Volume 13, Article 1922. 2022
- ◇ *Beyond exploitation: Measuring the impact of local search in swarm-based memetic algorithms through the interactions of individuals in the population*  
C. Santana, M. Oliveira, C. Bastos-Filho, R. Menezes. Swarm and Evolutionary Computation, 101040. 2022
- ◇ *Differences in the spatial landscape of urban mobility: gender and socioeconomic perspectives*  
M. Macedo, L. Lotero, A. Cardillo, R. Menezes, H. Barbosa. PLoS One. <https://doi.org/10.1371/journal.pone.0260874>. 2022
- ◇ *Reaching to the top: The gender effect in highly-ranked academics in computer science*  
A. Jaramillo, M. Macedo, R. Menezes. Advances in Complex Systems, Vol. 24, No. 03n04, 2150008. 2021
- ◇ *Breast cancer diagnosis using thermal image analysis: A data-driven approach based on swarm intelligence and supervised learning for optimized feature selection*  
M. Macedo, M. Santana, W. P. dos Santos, R. Menezes, C. Bastos-Filho. Applied Soft Computing, 107533. 2021
- ◇ *Uncovering the social interaction network in swarm intelligence algorithms*  
M. Oliveira, D. Pinheiro, M. Macedo, C. Bastos-Filho, R. Menezes. Applied Network Science 5, 1–20. 2020
- ◇ *Uncovering the differences and similarities between physical and virtual mobility*  
S. Hazarie, H. Barbosa, A. Frank, R. Menezes, G. Ghoshal. Journal of the Royal Society Interface 17 (168), 20200250. 2020
- ◇ *A data-driven social network intervention for improving organ donation awareness among minorities: analysis and optimization of a cross-sectional study*  
M. D. Murphy, D. Pinheiro, R. Iyengar, G. Lim, R. Menezes, M. Cadeiras. Journal of Medical Internet Research 22 (1), e14605. 2020
- ◇ *A data-driven network approach for characterization of political parties' ideology dynamics*  
J. Faustino, H. Barbosa, E. Ribeiro, R. Menezes. Applied Network Science 4 (1), 48. 2019

- ◇ *An approach for thematic relevance analysis applied to textual contributions in discussion forums*  
C. J. R. Machado, A. M. A. Maciel, R. L. Rodrigues, R. Menezes. International Journal of Distance Education Technologies (IJDET) 17 (3), 37–51. 2019
- ◇ *A data-driven framework for monitoring and improving organ donation awareness among minorities using optimal social network intervention*  
D. Pinheiro, M. Murphy, R. Iyengar, J. Lazareff, J. Choi, R. Menezes, M. Cadeiras. The Journal of Heart and Lung Transplantation 38 (4), S27. 2019
- ◇ *Unveiling the interplay between the TLR4/MD2 complex and HSP70 in the human cardiovascular system: a computational approach*  
A. A. de Oliveira, J. Faustino, M. E. de Lima, R. Menezes, K. P. Nunes. International Journal of Molecular Sciences 20 (13), 3121. 2019
- ◇ *Spatio-temporal variations in the urban rhythm: the travelling waves of crime*  
M. Oliveira, E. Ribeiro, C. Bastos-Filho, R. Menezes. EPJ Data Science 7 (1), 29. 2018
- ◇ *Towards evidence of long-range correlations in shallow seismic activities*  
D. Ferreira, J. Ribeiro, A. Papa, R. Menezes. EPL (Europhysics Letters) 121 (5), 58003. 2018
- ◇ *Human mobility: Models and applications*  
H. Barbosa-Filho, M. Barthelemy, G. Ghoshal, C. R. James, M. Lenormand, T. Louail, R. Menezes, J. J. Ramasco, F. Simini, M. Tomasini. Physics Reports 734, 1–74. 2018
- ◇ *The scaling of crime concentration in cities*  
M. Oliveira, C. Bastos-Filho, R. Menezes. PLoS One 12 (8), e0183110. 2017
- ◇ *On the effect of human mobility to the design of metropolitan mobile opportunistic networks of sensors*  
M. Tomasini, B. Mahmood, F. Zambonelli, A. Brayner, R. Menezes. Pervasive and Mobile Computing 38, 215–232. 2017
- ◇ *The role of human relations and interactions in designing memory-related models for sensor networks*  
B. Mahmood, R. Menezes. Sensors & Transducers 199 (4), 42. 2016
- ◇ *The effect of recency to human mobility*  
H. Barbosa, F. B. Lima-Neto, A. Evsukoff, R. Menezes. EPJ Data Science 4:21. 2015
- ◇ *Social-driven information dissemination for mobile wireless sensor networks*  
B. Mahmood, M. Tomasini, R. Menezes. Sensors and Transducers Journal, Volume 189, Issue 6. 2015
- ◇ *On the agreement between small-world-like OFC model and real earthquakes*  
D. S. R. Ferreira, A. Papa, R. Menezes. Physics Letters A, Volume 379, Issue 7. 2015
- ◇ *Using network science to assess particle swarm optimizers*  
M. Oliveira, C. J. A. Bastos-Filho, R. Menezes. Social Network Analysis and Mining 5:3. 2015
- ◇ *Using social media as a method of increasing organ donation*  
M. Cadeiras, M. Soroya, S. Patel, C. Starling, G. Lim, K. Dong, R. Menezes, M. Deng. Circulation. Suppl. 2. 2014
- ◇ *Small world picture of worldwide seismic events*  
D. S. R. Ferreira, A. Papa, R. Menezes. Physica A, Volume 408, pp. 170–180. 2014
- ◇ *A self-organized approach for detecting communities in networks*  
B. Collingsworth and R. Menezes. Social Network Analysis and Mining 4:169. 2014
- ◇ *Understanding the spread of malicious mobile-phone programs and their damage potential*  
P. Wang, M. C. González, R. Menezes, A.-L. Barabási. International Journal of Information Security 12 (5): 383–392. 2013
- ◇ *The United States heart and lung transplant “social” network: insights from the UNOS database*



- S. Venugopal, E. Stoner, M. Deng, R. Menezes, M. Cadeiras. *Journal of Heart and Lung Transplantation*, Vol. 32, Issue 4, Supplement, S159–S160. 2013
- ◇ *Understanding organ transplantation in the USA using geographical social networks*  
S. Venugopal, E. Stoner, M. Cadeiras, R. Menezes. *Social Network Analysis and Mining* 3 (3): 457–473. 2013
  - ◇ *A bio-inspired crime simulation model*  
V. Furtado, A. Melo, A. L. V. Coelho, R. Menezes, R. Perrone. *Decision Support Systems* 48 (1): 282–292. 2009
  - ◇ *Knowledge tuples never die, they just fade away*  
L. Nixon, R. Tolksdorf, R. Menezes, A. Wood. *Ubiquitous Computing and Communication Journal*, Vol. 3. 2008
  - ◇ *An adaptive in-network aggregation operator for query processing in wireless sensor networks*  
A. Brayner, A. Lopes, D. Meira, R. Vasconcelos, R. Menezes. *Journal of Systems and Software*, Volume 81, Issue 3, pp. 328–342. 2008
  - ◇ *Intelligent systems at Florida Tech*  
P. K. Chan, R. Menezes, D. Mitra, E. Ribeiro, M. Silaghi. *IEEE Intelligent Informatics Bulletin* 8 (1): 5–6. 2007
  - ◇ *Toward adaptive query processing in wireless sensor networks*  
A. Brayner, A. Lopes, D. Meira, R. Vasconcelos, R. Menezes. *Signal Processing*, Volume 87, No. 12, pp. 2911–2933. 2007
  - ◇ *Case studies for self-organization in computer science*  
M. Mamei, R. Menezes, R. Tolksdorf, F. Zambonelli. *Journal of Systems Architecture* 53 (8-9), pp. 443–460. 2006
  - ◇ *On coordination and its significance to distributed and multi-agent systems*  
S. Ossowski and R. Menezes. *Concurrency and Computation: Practice and Experience*. 2005
  - ◇ *On the semantics of coordination models for distributed systems: The LogOp case study*  
R. Menezes, A. Omicini, M. Viroli. *Electronic Notes in Theoretical Computer Science*, Volume 97, pp. 97–124. 2004
  - ◇ *Resource management in open Linda systems*  
R. Menezes. *Concurrency and Computation: Practice and Experience*, Vol. 15, pp. 1233–1256. 2003
  - ◇ *Coordination in a content-addressable web*  
R. Menezes, I. Merrick, A. Wood. *Autonomous Agents and Multi-Agent Systems Journal*, Vol. 2, No. 3, pp. 287–301. 1999

#### PEER-REVIEWED CONFERENCES AND WORKSHOPS

- ◇ *Network-centric analysis of memetic operators and communication topologies for swarm intelligence algorithms*  
C. Silva, C. Santana, C. Bastos-Filho, R. Menezes. *International Conference on Complex Networks*, 27–40. 2025
- ◇ *Understanding the structure and resilience of the Brazilian federal road network through network science*  
J. Taveira, F. B. L. Neto, R. Menezes. *International Conference on Complex Networks*, 107–121. 2025
- ◇ *The rephrased reality: Analysing sentiment shifts in LLM-rephrased text*  
F. L. Richardson, A. Villavicencio, R. Menezes. *Proceedings of the 40th ACM/SIGAPP Symposium on Applied Computing*, 920–927. 2025

- ◇ *Academic networks and resilience: An education perspective*  
P. Velasquez-Ferrufino, R. Menezes. Complex Networks & Their Applications XIII. Istanbul, Turkey. 2024
- ◇ *Network-based analysis of crossover operators in genetic algorithms*  
R. Rosenbaum, R. Menezes, C. Santana. Complex Networks & Their Applications XIII. Istanbul, Turkey. 2024
- ◇ *Branching out: How academic genealogy networks shape academic groups*  
G. Moraes, A. Brayner, R. Menezes. Complex Networks & Their Applications XIII. Istanbul, Turkey. 2024
- ◇ *Multidimensional analysis of the influence of socioeconomic and political indicators on the spread of COVID-19: A case study of Brazilian cities (2020–2024)*  
R. Reis, A. Brayner, M. Ângelo, R. Menezes. Simpósio Brasileiro de Banco de Dados (SBBD), 394–405. 2024
- ◇ *Public transit inequality in the context of the built environment*  
N. Iyer, R. Menezes, H. Barbosa. International Conference on Complex Networks, 203–218. 2024
- ◇ *Social interactions matter: Is grey wolf optimizer a particle swarm optimization variation?*  
R. C. Lira, M. Macedo, H. V. Siqueira, R. Menezes, C. Bastos-Filho. International Conference on Complex Networks, 101–114. 2024
- ◇ *Nature vs. nurture in science: The effect of researchers segregation on papers' citation histories*  
A. M. Jaramillo, F. Montes, R. Menezes. Proceedings of the 14th Conference on Complex Networks (CompleNet), 141–154. 2023
- ◇ *Network entropy as a measure of socioeconomic segregation in residential and employment landscapes*  
N. Iyer, R. Menezes, H. Barbosa. Proceedings of the 14th Conference on Complex Networks (CompleNet), 26–37. 2023
- ◇ *Academic mobility as a driver of productivity: a gender-centric approach*  
M. Macedo, A. M. Jaramillo, R. Menezes. Proceedings of the 14th Conference on Complex Networks (CompleNet), 120–131. 2023
- ◇ *Using vector fields in the modelling of movements as flows*  
S. Farokhnejad, M. Oliveira, E. Ribeiro, R. Menezes. Proceedings of the 14th Conference on Complex Networks (CompleNet), 155–167. 2023
- ◇ *A data-driven approach to cattle epidemic modelling under uncertainty*  
S. Farokhnejad, D. Cardoso, C. Rocha, A. S. da Mata, R. Menezes. Proceedings of the 13th Conference on Complex Networks (CompleNet). 2022
- ◇ *Networks of evolution: Modelling and deconstructing genetic algorithms using dynamic networks*  
C. Santana, E. Keedwell, R. Menezes. Proceedings of the Genetic and Evolutionary Computation Conference. 2022
- ◇ *Modelling the social interactions in grey wolf optimizer*  
R. C. Lira, M. Macedo, H. V. Siqueira, R. Menezes, C. Bastos-Filho. IEEE Latin American Conference on Computational Intelligence (LA-CCI). 2021
- ◇ *Fishing for interactions: A network science approach to modeling fish school search*  
M. Macedo, L. Taw, N. Gurrapadi, R. C. Lira, D. Pinheiro, M. Oliveira, R. Menezes. Proceedings of the Genetic and Evolutionary Computation Conference, 40–48. 2021
- ◇ *An approach to assess swarm intelligence algorithms based on complex networks*  
C. Santana, E. Keedwell, R. Menezes. Proceedings of the 2020 Genetic and Evolutionary Computation Conference, 31–39. 2020
- ◇ *A longitudinal analysis of vocabulary changes in social media*



- H. Hamoodat, F. Aswad, E. Ribeiro, R. Menezes. *Complex Networks XI*, 212–221. 2020
- ◇ *Communities of human migration in social media: An experiment in social sensing*  
F. Aswad, H. Hamoodat, E. Ribeiro, R. Menezes. *Complex Networks XI*, 222–232. 2020
  - ◇ *Characterizing the dynamics of academic affiliations: A network science approach*  
J. Faustino, N. Iyer, J. Mendonza, R. Menezes. *Complex Networks XI*, 393–404. 2020
  - ◇ *Network-based delineation of health service areas: A comparative analysis of community detection algorithms*  
D. Pinheiro, R. Hartman, E. Romero, R. Menezes, M. Cadeiras. *Complex Networks XI*, 359–370. 2020
  - ◇ *Gender patterns of human mobility in Colombia: Reexamining Ravenstein’s laws of migration*  
M. Macedo, L. Lotero, A. Cardillo, H. Barbosa, R. Menezes. *Complex Networks XI*, 269–281. 2020
  - ◇ *Short text tagging using nested stochastic block model: A Yelp case study*  
J. Bowllan, K. Cozart, S. M. M. Seyednezhad, A. Smith, R. Menezes. *International Conference on Complex Networks and Their Applications*, 822–833. 2019
  - ◇ *Modelling the social interactions in ant colony optimization*  
N. Gurrapadi, L. Taw, M. Macedo, M. Oliveira, D. Pinheiro, C. Bastos-Filho, R. Menezes. *International Conference on Intelligent Data Engineering and Automated Learning*, 216–222. 2019
  - ◇ *Characterizing the social interactions in the artificial bee colony algorithm*  
L. Taw, N. Gurrapadi, M. Macedo, M. Oliveira, D. Pinheiro, C. Bastos-Filho, R. Menezes. *IEEE Congress on Evolutionary Computation (CEC)*, 1243–1250. 2019
  - ◇ *Social media vocabulary reveals education attainment of populations*  
H. Hamoodat, E. Ribeiro, R. Menezes. *International Workshop on Complex Networks*, 157–168. 2019
  - ◇ *Entropy in network community as an indicator of language structure in emoji usage: A Twitter study across various thematic datasets*  
R. Hartman, S. M. M. Seyednezhad, D. Pinheiro, J. Faustino, R. Menezes. *International Conference on Complex Networks and Their Applications*, 328–337. 2018
  - ◇ *Author attribution using network motifs*  
Y. Al Rozz, R. Menezes. *International Workshop on Complex Networks*, 199–207. 2018
  - ◇ *Complex networks reveal a glottochronological classification of natural languages*  
H. Hamoodat, Y. Al-Rozz, R. Menezes. *International Workshop on Complex Networks*, 209–219. 2018
  - ◇ *Análise de Relevância Temática de Postagens em Fóruns de Discussão em Relação ao uso de Vídeos como Recurso Didático (in Portuguese)*  
C. Machado, A. Maciel, R. Rodrigues, R. Menezes. *Brazilian Symposium on Computers in Education*. 2018
  - ◇ *Improved multi-objective binary fish school for feature selection*  
M. Macedo, C. Bastos-Filho, R. Menezes. *FLAIRS*. 2018
  - ◇ *Refugee and immigration: Twitter as a proxy for reality*  
F. Aswad, R. Menezes. *FLAIRS*. 2018
  - ◇ *Characterization of users by using hourly and daily spatio-temporal patterns extracted from GPS trajectories*  
M. Tomasini, C. Bastos-Filho, R. Menezes. *FLAIRS*. 2018
  - ◇ *Emoji-word network analysis: Sentiments and semantics*  
S. M. M. Seyednezhad, H. Fede, I. Herrera, R. Menezes. *FLAIRS*. 2018

- ◇ *On the performance of network science metrics as long-term investment strategies in stock markets*  
A. Leone, M. Tomasini, Y. Al Rozz, R. Menezes. International Workshop on Complex Networks and Their Applications, 1053–1064. 2017
- ◇ *Representing emoji usage using directed networks: A Twitter case study*  
H. Fede, I. Herrera, S. M. M. Seyednezhad, R. Menezes. International Workshop on Complex Networks and Their Applications, 829–842. 2017
- ◇ *Assessing the suitability of network community detection to available meta-data using rank stability*  
R. Hartman, J. Faustino, D. Pinheiro, R. Menezes. Proceedings of the International Conference on Web Intelligence, 162–169. 2017
- ◇ *Language as a measure of welfare*  
P. Saha, D. Pacheco, R. Menezes. 30th Florida Artificial Intelligence Research Society Conference. 2017
- ◇ *Characterizing organ donation awareness from social media*  
D. F. Pacheco, D. Pinheiro, M. Cadeiras, R. Menezes. IEEE 33rd International Conference on Data Engineering (ICDE), 1541–1548. 2017
- ◇ *Characterization of written languages using structural features from common corpora*  
Y. Al Rozz, H. Hamoodat, R. Menezes. Workshop on Complex Networks CompleNet, 161–173. 2017
- ◇ *Understanding subject-based emoji usage using network science*  
S. M. M. Seyednezhad, R. Menezes. Workshop on Complex Networks CompleNet, 151–159. 2017
- ◇ *Automated robust anuran classification by extracting elliptical feature pairs from audio spectrograms*  
M. Tomasini, K. Smart, R. Menezes, M. Bush, E. Ribeiro. Acoustics, Speech and Signal Processing (ICASSP). 2017
- ◇ *A language-centric study of Twitter connectivity*  
P. Saha, R. Menezes. International Conference on Social Informatics, 485–499. 2016
- ◇ *Exploring the world languages in Twitter*  
P. Saha, R. Menezes. IEEE/WIC/ACM International Conference on Web Intelligence (WI), 153–160. 2016
- ◇ *Characterization of football supporters from Twitter conversations*  
D. F. Pacheco, D. Pinheiro, F. B. de Lima-Neto, E. Ribeiro, R. Menezes. IEEE/WIC/ACM International Conference on Web Intelligence (WI), 169–176. 2016
- ◇ *Weighted multi-resource minority games*  
S. M. M. Seyednezhad, E. Shinseki, D. Romero, R. Menezes. Proceedings of SAI Intelligent Systems Conference, 285–305. 2016
- ◇ *Communication diversity in particle swarm optimizers*  
M. Oliveira, D. Pinheiro, B. Andrade, C. Bastos-Filho, R. Menezes. International Conference on Swarm Intelligence, 77–88. 2016
- ◇ *A data science approach for quantifying spatio-temporal effects to graft failures in organ transplantation*  
D. Pinheiro, F. Hamad, M. Cadeiras, R. Menezes, N. Nezamoddini-Kachouie. IEEE 38th Annual International Conference of the Engineering in Medicine and Biology Society (EMBC), 3433–3436. 2016
- ◇ *Multi-resource minority games: Redefining the game*  
D. Romero, E. Shinseki, S. M. M. Seyednezhad, R. Menezes. IntelliSys 2016. London.
- ◇ *Extracting health-related social structures from conversations in Twitter*  
A. Al-Rubaye, R. Menezes. ACM Hypertext 2016.

- ◇ *A social-based strategy for memory management in sensor networks*  
B. Mahmood, R. Menezes. SensorNets. Rome. 2016
- ◇ *Returners and explorers dichotomy in web browsing behavior: A human mobility approach*  
H. Barbosa, F. B. de Lima Neto, A. Evsukoff, R. Menezes. In Complex Networks VII (pp. 173–184). Springer Verlag. 2016
- ◇ *Estimating memory requirements in wireless sensor networks using social tie strengths*  
B. Mahmood, M. Tomasini, R. Menezes. IEEE SenseAPP. Florida, USA. 2015
- ◇ *Football conversations: What Twitter reveals about the 2014 World Cup*  
D. Pacheco, F. B. de Lima-Neto, R. Menezes. BraSNAM 2015. Recife, Brazil. 2015
- ◇ *From criminal spheres of familiarity to crime networks*  
M. Oliveira, H. Barbosa-Filho, T. Yehle, S. White, R. Menezes. CompleNet. New York. 2015
- ◇ *Social-based forwarding of messages in sensor networks*  
B. Mahmood, M. Tomasini, R. Menezes. SensorNets. France. 2015
- ◇ *The spatial structure of crime in urban environments*  
S. White, T. Yehle, H. Serrano, M. Oliveira, R. Menezes. CrimeNet (at SocInfo). Spain. 2014
- ◇ *WhatFrog: Development of app and web interface for automated anuran recognition and mapping*  
K. Smart, H. Serrano, M. Luce, M. B. Bush, R. Menezes, E. Ribeiro. 99th ESA Annual Meeting. 2014
- ◇ *Using interactions in the quantification of media bias*  
D. F. Pacheco, D. Rose, F. B. Lima-Neto, R. Menezes. SocialCom. Stanford. 2014
- ◇ *Political social networks reveal strong party loyalty in Brazil and weak regionalism*  
M. Oliveira, C. J. A. Bastos-Filho, R. Menezes. SocialCom. Stanford. 2014
- ◇ *Towards a network-based approach to analyze particle swarm optimizers*  
M. Oliveira, C. J. A. Bastos-Filho, R. Menezes. IEEE Symposium on Swarm Intelligence (SIS), 1–8. 2014
- ◇ *Negative implications of a power-law distribution: A study on networks of scientific reviewers*  
S. Qin, M. C. Silaghi, R. Menezes, W. Cheung. Workshop on Complex Networks, Volume 549, 305–317. 2014
- ◇ *The small world of seismic events*  
D. S. R. Ferreira, A. Papa, R. Menezes. Workshop on Complex Networks, Volume 549, 97–105. 2014
- ◇ *Evaluating the performance of social networks of sensors under different mobility models*  
M. Tomasini, F. Zambonelli, A. Brayner, R. Menezes. ASE/IEEE International Conference on Social Computing (SocialCom). Washington, DC. 2013
- ◇ *Using patterns of social dynamics in the design of social networks of sensors*  
M. Tomasini, F. Zambonelli, R. Menezes. IEEE International Conference on Internet of Things, 685–692. Beijing, China. 2013
- ◇ *United States Congress relations according to liberal and conservative newspapers*  
B. Mahmood, R. Menezes. 2nd IEEE Network Science Workshop (NSW), 98–101. 2013
- ◇ *On the relation between tourism and trade: A network experiment*  
J. Brahmabhatt, R. Menezes. 2nd IEEE Network Science Workshop (NSW), 74–81. 2013
- ◇ *Understanding history through networks: The Brazil case study*  
H. S. Barbosa-Filho, F. B. de Lima-Neto, R. Menezes. Complex Networks IV, Studies in Computational Intelligence 476, 101–108. 2013

- ◇ *Assessing particle swarm optimizers using network science metrics*  
M. A. C. Oliveira-Júnior, C. J. A. Bastos-Filho, R. Menezes. Complex Networks IV, Studies in Computational Intelligence 476, 173–184. 2013
- ◇ *An analysis of the overlap of categories in a network of blogs*  
P. Saha, R. Menezes. Complex Networks IV, Studies in Computational Intelligence 476, 59–70. 2013
- ◇ *Strategies, political position, and electoral performance of Brazilian political parties* (Best Paper Award)  
H. Barbosa-Filho, J. Faustino, R. R. Martins, R. Menezes. BRICS-CCI Complex System Symposium. 2013
- ◇ *Using network sciences to evaluate the Brazilian airline network*  
D. Oliveira, M. Carvalho, R. Menezes. 19th International Conference Computing and Combinatorics (COCOON), 849–858. 2013
- ◇ *The social structure of organ transplantation in the United States*  
S. Venugopal, E. Stoner, M. Cadeiras, R. Menezes. Complex Networks, Studies in Computational Intelligence 424, 199–206. 2012
- ◇ *A framework for a model wireless sensor network*  
P. Martins, R. Menezes, I. Hidalgo, U. Fritzke. 9th European Conference on Wireless Sensor Networks, Trento. 2012
- ◇ *The effect of citations to collaboration networks*  
P. Divakarmurthy, R. Menezes. Complex Networks, Studies in Computational Intelligence 424, 177–185. 2012
- ◇ *Measuring a category-based blogosphere*  
P. Saha, R. Menezes. Complex Networks, Studies in Computational Intelligence 424, 131–139. 2012
- ◇ *Using network science to define a dynamic communication topology for particle swarm optimizers*  
M. A. C. Oliveira-Junior, C. J. A. Bastos-Filho, R. Menezes. Complex Networks, Studies in Computational Intelligence 424, 39–47. 2012
- ◇ *SOCIAL: A self-organized entropy-based algorithm for identifying communities in networks*  
B. Collingsworth, R. Menezes. 6th IEEE International Conference on Self-Adaptive and Self-Organizing Systems, 217–222. 2012
- ◇ *Dynamic composition of coordination abstractions for pervasive systems: the case of LogOp*  
R. Menezes, A. Omicini, M. Viroli. ACM Symposium on Applied Computing (SAC), 1557–1559. 2012
- ◇ *A self-organized approach for detecting communities in networks*  
B. Collingsworth, R. Menezes. Intelligent Distributed Computing VI, Studies in Computational Intelligence 446, 29–39. 2012
- ◇ *Area diversity in computer science collaborations*  
P. Divakarmurthy, R. Menezes. ACM Symposium on Applied Computing (SAC), 2041–2042. 2012
- ◇ *A temporal analysis of geographical distances in computer science collaborations*  
P. Divakarmurthy, P. Biswas, R. Menezes. IEEE International Conference on Social Computing, 657–660. 2011
- ◇ *Using network science to understand the structure of Brazilian popular music*  
C. Gunaratna, R. Menezes. 11th IEEE International Conference on Computer and Information Technology (CIT), 395–402. 2011
- ◇ *Using network sciences to rank musicians and composers in Brazilian popular music*  
C. Gunaratna, E. Stoner, R. Menezes. International Symposium/Conference on Music Information

Retrieval, 441–446. 2011

- ◇ *A network-centric epidemic approach for automated image label annotation*  
M. Bohlool, R. Menezes, E. Ribeiro. Communications in Computer and Information Science 116, 138–145. 2011
- ◇ *Using networks to understand the dynamics of software development*  
C. Roach, R. Menezes. Communications in Computer and Information Science 116, 119–129. 2011
- ◇ *Using network analysis to understand the relation between cuisine and culture*  
D. K. Kular, R. Menezes, E. Ribeiro. IEEE Workshop on Network Science (NSW). West Point, NY, 38–45. 2011
- ◇ *The dynamics of the MAPK network*  
I. Policeanu, R. Menezes. 9th International Workshop on Data Mining in Bioinformatics (BIOKDD). Washington, DC. 2010
- ◇ *A recommender system for YouTube based on its network of reviewers*  
S. Qin, R. Menezes, M. Silaghi. 2nd IEEE International Conference on Social Computing, 323–328. 2010
- ◇ *Self organization in coordination systems using a WordNet-based ontology*  
D. Pianini, S. Virruso, R. Menezes, A. Omicini, M. Viroli. 4th IEEE International Conference on Self-Adaptive and Self-Organizing Systems, 114–123. 2010
- ◇ *Hybrid and evolutionary agent-based social simulations using Lévy flight as randomization method*  
F. B. de Lima Neto, R. Menezes, M. Álvares. IEEE Congress on Evolutionary Computation, 1–8. 2010
- ◇ *Evaluating failure in terrain coverage by autonomous agents*  
R. Treverton, R. Menezes. IEEE Workshop on Robotic Intelligence in Informationally Structured Space, 79–86. 2009
- ◇ *Assessing organizational stability via network analysis*  
B. Collingsworth, R. Menezes, P. Martins. IEEE Symposium on Computational Intelligence for Financial Engineering, 43–50. 2009
- ◇ *Self-organized control of knowledge generation in pervasive computing systems*  
G. Castelli, R. Menezes, F. Zambonelli. ACM Symposium on Applied Computing, 1202–1208. 2009
- ◇ *Identification of social tension in organizational networks: Relating clan/clique formation to social tension*  
B. Collingsworth, R. Menezes. Studies in Computational Intelligence 207, 209–223. 2009
- ◇ *Autonomous algorithms for terrain coverage: Metrics, classification and evaluation*  
J. Mason, R. Menezes. IEEE Congress on Evolutionary Computation, 1641–1648. 2008
- ◇ *On the performance of swarm-based tuple organization in Linda systems*  
D. Graff, R. Menezes, R. Tolksdorf. IEEE Congress on Evolutionary Computation, 2709–2716. 2008
- ◇ *A study of terrain coverage models*  
R. Menezes, H. Bullen. ACM Symposium on Applied Computing (SAC), 1964–1968. 2008
- ◇ *Handling dynamic networks using evolution in ant-colony optimization*  
C. Roach, R. Menezes. New Frontiers in Applied Artificial Intelligence, LNCS 5027, 795–804. 2008
- ◇ *Models for temporal and spatial terrain coverage*  
H. Bullen, R. Menezes. International Symposium on Intelligent and Distributed Computing (IDC 2007). Studies in Computational Intelligence 78, 239–244. 2008
- ◇ *Using ant’s alarm pheromone to improve software testing automation*  
R. Menezes, R. Silva, M. Barros, A. M. Silva. IDC 2007. Studies in Computational Intelligence 78,

115–124. 2008

- ◇ *A simulator for query processing in wireless sensor networks (in Portuguese)*  
D. Meira, A. Brayner, R. Menezes, A. Lopes. 2007 Brazilian Symposium on Databases (SBBD 2007). João Pessoa. 2007
- ◇ *Self-organized over-clustering avoidance in tuple-space systems*  
M. Casadei, R. Menezes, M. Viroli, R. Tolksdorf. IEEE Congress on Evolutionary Computation. Singapore, 1408–1415. 2007
- ◇ *Using ant's brood sorting to increase fault tolerance in Linda's tuple distribution mechanism*  
M. Casadei, R. Menezes, M. Viroli, R. Tolksdorf. 11th International Workshop on Cooperative Information Agents. Delft, Netherlands. LNCS 4676, 255–269. 2007
- ◇ *On the problem of over-clustering in tuple-based coordination systems*  
M. Casadei, R. Menezes, M. Viroli, R. Tolksdorf. 1st IEEE International Conference on Self-Adaptive and Self-Organizing Systems (SASO). Boston. 2007
- ◇ *Coordinating knowledge in pervasive environments (Best Paper Award)*  
L. Nixon, R. Tolksdorf, A. Wood, R. Menezes. Workshop on CoMA at WETICE. Paris. 2007
- ◇ *Network distribution of security policies via ant-like foraging behavior*  
A. Ondi, R. Menezes, R. Ford. International Conference on Internet and Web Applications and Services. Mauritius. 2007
- ◇ *A model for terrain coverage inspired by ant's alarm pheromones*  
R. Menezes, F. Martins, F. E. Vieira, R. Silva, M. Braga. ACM Symposium on Applied Computing (SAC). Seoul. 728–732. 2007
- ◇ *Balancing energy consumption and memory usage in sensor data processing*  
A. Brayner, A. Lopes, R. Vasconcelos, R. Menezes. ACM Symposium on Applied Computing (SAC). Seoul. 935–939. 2007
- ◇ *A crime simulation model based on swarm intelligence*  
V. Furtado, A. Melo, A. Coelho, R. Menezes. ACM Symposium on Applied Computing (SAC). Seoul. 56–57. 2007
- ◇ *ADAGA – Adaptive aggregation algorithm for sensor networks*  
A. Brayner, A. Lopes, D. Meira, R. Vasconcelos, R. Menezes. 2006 Brazilian Symposium on Databases (SBBD 2006). Florianópolis. 2006
- ◇ *Evolution in swarm intelligence: An evolutionary ant-based optimization algorithm*  
C. Roach, R. Menezes. International Workshop on Ant-Colony Optimization and Swarm Intelligence (ANTS 2006). Brussels. 2006
- ◇ *Self-organized and social models of criminal activity in urban environments*  
A. Melo, R. Menezes, V. Furtado, A. Coelho. International Workshop on Ant-Colony Optimization and Swarm Intelligence (ANTS 2006). Brussels. 2006
- ◇ *On the idea of using nature-inspired metaphors to improve software testing*  
F. E. Vieira, F. Martins, R. Silva, R. Menezes, M. Braga. 3rd IFIP Conference on Artificial Intelligence Applications and Innovations (AIAI 2006). Athens. 2006
- ◇ *Increasing sales in supermarkets via real-time information about customer's activities – the swarm-moves simulation*  
Z. Usmani, R. Menezes. International Conference on Modeling, Simulation and Visualization Methods. Las Vegas. 2006
- ◇ *Using self-organization in an agent framework to model criminal activity in response to police patrol routes*  
V. Furtado, A. Melo, R. Menezes, M. Belchior. Florida Artificial Intelligence Research Society



International Conference (FLAIRS). 2006

- ◇ *The fading concept in tuple-space systems*  
R. Menezes, A. Wood. ACM Symposium on Applied Computing (SAC). Dijon. 2006
- ◇ *A perspective on the use of self-organization in tuple-space coordination systems*  
R. Menezes. International Conference on Artificial Intelligence (ICAI 05). Las Vegas. 2005
- ◇ *Swarming computer security: An experiment in policy distribution*  
R. Menezes, R. Ford, A. Ondi. IEEE Swarm Intelligence Symposium (SIS). Pasadena. 2005
- ◇ *Self-organization and computer security: A case study in adaptive coordination*  
R. Menezes. ACM Symposium on Applied Computing (SAC). Santa Fe. 2005
- ◇ *The role of social structures in mobile ad-hoc networks*  
P. Dhakan, R. Menezes. 43rd ACM Southeast Regional Conference, 59–64. 2005
- ◇ *Simulating robot collective behavior using StarLogo*  
A. Tinkham, R. Menezes. 2004 ACM Southeastern Conference. Huntsville, 396–401. 2004
- ◇ *On the implementation of SwarmLinda*  
A. Charles, R. Menezes, R. Tolksdorf. 2004 ACM Southeastern Conference. Huntsville, 297–298. 2004
- ◇ *Adaptiveness in Linda-based coordination models*  
R. Menezes, R. Tolksdorf. 1st International Workshop on Engineering Self-Organizing Applications (ESOA). Melbourne, 212–232. 2003
- ◇ *On the implementation of an architecture for distributed control systems*  
X. Shen, H. Gutierrez, R. Menezes. 7th World Multiconference on Systemics, Cybernetics and Informatics. Orlando. 2003
- ◇ *Using swarm intelligence in Linda systems*  
R. Tolksdorf, R. Menezes. 4th International Workshop on Engineering Societies in the Agents World (ESAW). London. LNAI 3071. 2003
- ◇ *A new approach to scalable Linda-systems based on swarms*  
R. Menezes, R. Tolksdorf. ACM Symposium on Applied Computing (SAC), 375–379. Melbourne. 2003
- ◇ *Coordination of distributed components using LogOp*  
R. Menezes, J. Snyder. International Conference on Parallel and Distributed Processing Techniques and Applications, 109–114. 2003
- ◇ *Have ReSpeCT for LogOp*  
R. Menezes, M. Viroli, A. Omicini. 2002 Italian Workshop on Objects and Agents (WOA). Milan. 2002
- ◇ *A coordination architecture for distributed autonomous control systems*  
X. Shen, R. Menezes, H. Gutierrez. International Conference on Parallel and Distributed Processing (PDPTA). Las Vegas. 2002
- ◇ *Using logical operators as an extended coordination mechanism in Linda*  
J. Snyder, R. Menezes. Coordination 2002. York.
- ◇ *Experience with memory management in open Linda systems*  
R. Menezes. ACM Symposium on Applied Computing (SAC). Las Vegas. 2001
- ◇ *On the introduction of location in Linda systems*  
R. Menezes. 4th World Multiconference on Systemics, Cybernetics and Informatics. Orlando.
- ◇ *Distributed garbage collection of tuple spaces in open Linda coordination systems*

R. Menezes, A. Wood. 15th International Symposium on Computer and Information Sciences (ISCIS). Kupadasý, Turkey. 1999

- ◇ *Coordination of distributed I/O in tuple space systems*  
R. Menezes, A. Wood. 31st IEEE Hawaii International Conference on System Sciences (HICSS). Big Island. 1999
- ◇ *Using tuple monitoring and process registration on the implementation of garbage collection in open Linda-like systems*  
R. Menezes, A. Wood. 10th IASTED International Conference (PDCS). Las Vegas. 1998
- ◇ *Ligia: Incorporating garbage collection in a Java based Linda-like run-time system*  
R. Menezes. 2nd Workshop on Distributed Systems (WOSID). Curitiba, Brazil. 1998
- ◇ *Garbage collection in open distributed tuple space systems*  
R. Menezes, A. Wood. 15th Brazilian Computer Network Symposium (SBRC). São Carlos, Brazil. 1997

#### PEER-REVIEWED PRESENTATIONS (WITHOUT PUBLICATIONS)

- ◇ *Learning complex network structures through graph-augmented normalizing flows for enhanced anomaly detection in multivariate time series*  
C. Silva, A. M. Ribeiro, R. Monteiro, R. Menezes, C. Bastos-Filho, D. Pinheiro. XVI International Conference on Complex Networks. 2025
- ◇ *Revealing swarm intelligence patterns using network science*  
R. C. L. Silva, M. Macedo, H. Siqueira, R. Menezes, C. Bastos-Filho. XVI International Conference on Complex Networks. 2025
- ◇ *Statistics on Reddit: entropy and Gini coefficient*  
G. Machado, G. Baxter, D. Pacheco, R. Menezes. NetSci. 2023
- ◇ *Academic mobility as a driver of productivity: A gender-centric approach*  
M. Macedo, A. Jaramillo, R. Menezes. NetSci. 2023
- ◇ *Mobility and transit segregation in urban spaces*  
N. Iyer, R. Menezes, H. Barbosa. NetSci. 2023
- ◇ *Changes in the time-space dimension of human mobility during the COVID-19 pandemic*  
C. Santana, F. Botta, H. Barbosa, F. Privitera, R. Menezes, R. Di Clemente. NetSci. 2023
- ◇ *Gender representation and homophily as barriers to women's career in science*  
A. Jaramillo, M. Macedo, M. Oliveira, F. Karimi, R. Menezes. NetSci. 2023
- ◇ *Contrasting social and non-social sources of predictability in human mobility*  
Z. Chen, S. Kelty, A. Evsukoff, B. Foucault-Welles, J. Bagrow, R. Menezes, G. Ghoshal. 6th Northeast Regional Conference on Complex Systems. Potsdam, NY. 2023
- ◇ *From networks to flows: Using flow maps to understand mobility patterns in cattle trade*  
S. Farokhnejad, E. Ribeiro, R. Menezes. NetSci-X 2023. Buenos Aires. 2023
- ◇ *Minimizing degree centrality and connectedness as a long term portfolio investment strategy*  
A. Leone, M. Tomasini, Y. Al Rozz, R. Menezes. NetSci 2017. Indianapolis. 2017
- ◇ *Uncovering the role of spatial constraints in physical and virtual mobility*  
S. Hazarie, H. Barbosa, A. Frank, R. Menezes, G. Ghoshal. 3rd Northeast Regional Conference on Complex Systems. Online. 2020
- ◇ *Social network health interventions: Can Mayweather and Pacquiao "knock-out" health disparities?*  
M. Cadeiras, C. Starling, S. Soroya, M. Bakir, E. Chang, G. Lim, T. Mone, M. Deng, R. Menezes. American Heart Association. 2015

- ◇ *Network-based assessment of swarm-based algorithms*  
D. Pinheiro, M. Oliveira, C. J. A. Bastos-Filho, R. Menezes. NetSci 2015. Zaragoza. 2015
- ◇ *Scaling laws of crime*  
M. Oliveira, R. Menezes. NetCrime at NetSci 2015. Zaragoza. 2015
- ◇ *Characterizing preferences and returns in human mobility*  
H. Barbosa, F. B. Lima-Neto, R. Menezes. NetMob 2015. Boston. 2015
- ◇ *Dynamics of fans borders during the 2014 FIFA World Cup*  
D. Pacheco, R. Menezes. NetSci-X 2015. Rio de Janeiro. 2015
- ◇ *Design of urban social networks of sensors*  
M. Tomasini, F. Zambonelli, A. Brayner, R. Menezes. NetSci 2014. Berkeley. 2014
- ◇ *Positions, strategies, and electoral performance: Analyzing Brazilian party switching networks*  
H. Barbosa-Filho, F. B. Lima-Neto, R. Menezes. NetSci 2014. Berkeley. 2014
- ◇ *Using networks to divide the USA into regions to improve the process of organ transplantations*  
S. Venugopal, E. Stoner, M. Cadeiras, R. Menezes. 5th Workshop on Complex Networks (CompleNet 2014). Bologna. 2014
- ◇ *Small world picture of worldwide seismic events*  
D. S. R. Ferreira, A. Papa, R. Menezes. Mathematical Modeling of Complex Systems. École Centrale Paris. 2013
- ◇ *Revealing the structure of online programming competitions*  
H. Miller, R. Menezes, K. Walrond-McClean. 3rd Workshop on Complex Networks (CompleNet 2012). Melbourne, Florida. 2012
- ◇ *Network sciences in music recommendation: A case study with Brazilian music*  
R. Menezes, C. Gunaratna, M. Patel. 3rd Workshop on Complex Networks (CompleNet 2012). Melbourne, Florida. 2012
- ◇ *Email network analysis as an early indicator of organization stability*  
R. Menezes, B. Collingsworth. NetSci 2010. Boston. 2010
- ◇ *Epidemics on mobile phone networks*  
P. Wang, M. C. González, R. Menezes, A.-L. Barabási. Workshop on Analysis of Mobile Phone Networks at NetSci 2010. Boston. 2010
- ◇ *Assortative mixing in the Amazon.com book network*  
B. Collingsworth, R. Menezes. 2nd Workshop on Complex Networks (CompleNet 2010). Rio de Janeiro. 2010

## EDITED JOURNALS (SPECIAL ISSUES)

- ◇ *Navigating information integrity in the age of misinformation*  
D. Pacheco, J. Bollen, R. Menezes. Applied Network Science. Springer Nature. 2025
- ◇ *Social system as complex networks*  
G. Ghoshal, G. Mangioni, R. Menezes, J. Poncela-Casanovas. Social Network Analysis and Mining 4 (1). 2014
- ◇ *Special issue on self-organizing coordination*  
M. Casadei, M. Mamei, R. Menezes. Science of Computer Programming, Volume 78. 2013
- ◇ *Special issue on intelligent distributed computing*  
C. Badica, R. Menezes. Concurrency and Computation: Practice and Experience 24 (6). 2012
- ◇ *Special issue on semantic extensions to middleware*  
R. Krummenacher, E. P. B. Simperl, R. Menezes. Journal of Software 4 (8). 2009
- ◇ *Special issue: Coordination models and systems*  
S. Ossowski, R. Menezes. Concurrency and Computation: Practice and Experience 18 (4). 2006
- ◇ *Special issue on nature-inspired applications and systems*  
M. Farooq, R. Menezes. Journal of Systems Architecture 52 (8-9). 2006
- ◇ *Special issue: Computer security*  
G. Bella, R. Menezes. Concurrency – Practice and Experience 16 (11). 2004

## TECHNICAL REPORTS

- ◇ *Towards evidences of long-range correlations in seismic activity*  
D. S. R. Ferreira, A. Papa, R. Menezes. arXiv:1405.0307. 2014
- ◇ *New generation of mobile phone viruses and corresponding countermeasures*  
P. Wang, M. C. González, R. Menezes, A.-L. Barabási. The Computing Research Repository. 2010
- ◇ *On the implementation of SwarmLinda: A Linda system based on swarm intelligence*  
A. Charles, R. Menezes, R. Tolksdorf. Florida Institute of Technology, Department of Computer Sciences, CS-2004-03. 2004
- ◇ *A new approach to scalable Linda-systems based on swarms (extended version)*  
R. Menezes, R. Tolksdorf. Florida Institute of Technology, Department of Computer Sciences, CS-2003-04. 2003
- ◇ *On the use of Linda as a framework for distributed database systems*  
M. Thirukonda, R. Menezes. Florida Institute of Technology, Department of Computer Sciences, CS-2002-02. 2002
- ◇ *Ligia: A Java based Linda-like run-time system with garbage collection of tuple spaces*  
R. Menezes, A. Wood. University of York, Department of Computer Science, YCS 304. 1998
- ◇ *Parallel computing models and algorithm design (in Portuguese)*  
R. Menezes, J. C. Setubal. Institute of Computing, State University of Campinas, Brazil. 1995

## EXHIBITIONS

---

*The Art of Networks II*, Poland, January 2016

*The Art of Networks II*, Foosaner Art Museum, September–November 2015

*The Art of Networks II*, New York Hall of Science, March–May 2015

*The Art of Networks*, Foosaner Art Museum, March–April 2012

*Analyse This: Digital Models of Swarm Intelligence*, Ruth Funk Center for Textile Arts, September 2010

## WORK EXPERIENCE

---

### ACADEMIC

#### Computer Science Director of Global Development

*University of Exeter, UK*

September 2024 – Present

#### Professor of Data and Network Science

*University of Exeter, UK*

August 2018 – Present

#### External Examiner

*University of Sussex, UK*

January 2025 – December 2025

#### Visiting Professor

*Federal University of Ceará, Brazil*

October 2022 – 2024

#### External Examiner

*University of Warwick, UK*

January 2021 – 2024

#### Alan Turing Fellow

*The Alan Turing Institute, UK*

October 2021 – January 2024

#### Head of the Computer Science Department

*University of Exeter, UK*

August 2018 – August 2022

#### Professor in Computer Science

*Florida Institute of Technology, USA*

March 2016 – 2018

Teaching Activities: Social Network Analysis, Complex Networks, Algorithms and Data Structures, Computing Careers and Disciplines.

Joint appointments: School of Human-Centered Design, Innovation & Art

#### Director of the School of Computing

*Florida Institute of Technology, USA*

December 2016 – October 2017

#### Department Head of Computer Science

*Florida Institute of Technology, USA*

December 2015 – December 2016

#### Associate Professor in Computer Science

*Florida Institute of Technology, USA*

August 2006 – 2016

Teaching Activities: Social Network Analysis, Complex Networks, Algorithms and Data Structures, Swarm Intelligence, Introduction to Biologically Inspired Computing.

#### Visiting Researcher

*Department of Physics, Northeastern University, CCNR (Barabasi Lab), Boston, USA* January 2009 –

July 2009

### Visiting Researcher

*IVIA Ltda., Brazil*

September 2005 – 2007

### Assistant Professor in Computer Science

*Florida Institute of Technology, USA*

August 2000 – 2006

Teaching Activities: Java, Distributed Systems, Network Programming, Algorithms and Data Structures, Coordination Systems, Swarm Intelligence, Introduction to Biologically Inspired Computing.

### Lecturer in Computer Science (Assistant Professor)

*University of Ulster, UK*

August 1999 – July 2000

Teaching Activities: Object Oriented Programming

### Teaching Assistant

*The University of York, UK*

August 1996 – July 1999

Teaching Activities: Theory of Computation, Pascal and Java Programming.

### Adjunct Faculty in Computer Science

*Padre Anchieta University, Brazil*

November 1994 – June 1995

Teaching Activities: Algorithms and Data Structures

## INDUSTRIAL

### Web Technology Consultant

*8over8 Limited, UK*

February 2000 – July 2000

### Systems Consultant

*Mota Machado Constructors, Brazil*

January 1993 – June 1995

### Computing Services Director

*Fiutex Textile Industry, Brazil*

July 1991 – February 1993

### Programmer / System Analyst

*Mota Machado Constructors, Brazil*

October 1989 – July 1991

## TEACHING ACTIVITIES

---

## GRADUATE COURSES

### Complex Networks

An advanced version of the Social Network Analysis course. Students produce a conference-quality research paper. The course includes heavier mathematical formalism and studies non-social networks (technological, biological, etc.)

### Social Networks Analysis

A seminar-like course on computational approaches to social networks covering modelling, data collection, and structural analysis.



## **Introduction to Biologically Inspired Computing**

An introductory course covering most areas of computer science inspired by biological systems.

## **Swarm Intelligence**

An advanced course for senior graduate students. One of the first courses in the area offered worldwide; material has been used at EPFL (Switzerland) and JPL/Caltech (USA).

## **Coordination Systems**

An advanced course on the theory and practice of coordination. The first course in the world on this topic, covering computation and coordination as distinct components of distributed systems.

## **Distributed Systems**

Standard course on distributed systems as a core area of computer science.

## **Network Programming**

A project-based course where students write programs using computer network concepts.

## **UNDERGRADUATE COURSES**

### **Fundamentals of Software Development II**

A freshman programming course reorganised to encourage diligent study with constant feedback. Includes 4 exams, 10 pop quizzes, 2 semester projects, and 15 lab exercises.

### **Algorithms and Data Structures**

A core CS curriculum course emphasising lab practice and continuous feedback.

## **SELECTED INVITED AND KEYNOTE TALKS**

Recipe/Food Networks as Windows into Inequality: A Call for Computational Analysis of Eating Behaviours. First International Workshop of Scientific and Computational Gastronomy. Parma, Italy. September 2025

Violence, City Size and Geographical Isolation in African Cities. LANET. Punta del Este, Uruguay. August 2025

Random Thoughts about Causality and Mobility. European Mobility Symposium. Pisa, Italy. November 2024

Uncovering the differences and similarities between physical and virtual mobility. Lipari School of Complex Systems. Italy. July 2023

Gender Inequality in Scientific Production. LANET. Cusco, Peru. August 2023

Disaggregating Data: A Case of Gender Differences in Collaboration Networks and Urban Human Mobility. Data Science Africa. Arusha, Tanzania. July 2022

Differences in the spatial landscape of urban mobility: Gender and socioeconomic perspectives. Alan Turing Institute. Urban Analytics. Leeds, UK. 2021

Network Science as a Framework for Understanding Population Health. One Health One Planet Meeting. Lavras, Minas Gerais, Brazil. 2021

Using Data Analytics to Understand Human Mobility Patterns in Cities. Analytics Forum. Colombia. 2020

The Next Big Thing in Big Data. CSoNet. Ho Chi Minh City. Vietnam. 2019

On the Structure, Concentration and Dynamics of Urban Crime. LA-CCI. Guayaquil, Ecuador. 2019

On the Structure, Concentration and Dynamics of Urban Crime. MARAMI. Dijon. France. 2019

On the Scale, Concentration and Dynamics of Crime. Urban Analytics Forum. Alan Turing Institute. Bristol. 2019

On the Scale, Concentration and Dynamics of Crime. Cambridge Networks Day. 2019

On the Scale, Concentration and Dynamics of Crime. LANET. Cartagena, Colombia. 2019

On the Scale, Concentration and Dynamics of Crime. Lipari School of Complex Systems. Italy. July 2019

Human Dynamics and Landscape Decisions: Why People and Urban Systems Matter. Isaac Newton Institute. University of Cambridge, UK. July 2019

On the Scale, Concentration and Dynamics of Crime. University of Aveiro, Portugal. July 2019

On the Scale, Concentration and Dynamics of Crime. Alan Turing Institute, London. June 2019

Crime Concentration in Cities. IFISC. Palma, Spain. 2019

Self-Organisation in Computer Science. Christmas Lecture, University of Exeter, UK. 2018

Spatio-Temporal Regularities in Crime Dynamics. Oxford Mathematics Institute, UK. 2018

Human Mobility. LA-CCI. Arequipa, Peru. 2017

The Scaling of Crime Concentration in Cities. LANET. Puebla, Mexico. 2017

Assessing Swarm System using Network Sciences. NetSci-X 2015. Rio de Janeiro. 2015

Human Mobility: Social Relations Applied to Smart Cities. Science for the City. Fortaleza, Brazil. October 2013

Using Swarm Intelligence and Network Sciences in Crime Modeling. Round table at BRICS-CCI. Recife, Brazil. September 2013

Evaluating the Performance of Social Networks of Sensors under Different Mobility Models. University of Fortaleza, Brazil. September 2013

Advances in Network Science fundamental research. How are we doing? Panel Discussion. IEEE Network Science Workshop. May 2013

Short Course on Network Science. AGH University of Science and Technology. Krakow, Poland. June 2012

Research at BioComplex Lab, Florida Tech. State University of Pernambuco. May 2011

Short Course on Complex Networks. University of Fortaleza, Brazil. August 2010

Short Course on Bio-Inspired Computing. University of Fortaleza, Brazil. June 2006

Trends in Information Technology. Unifor World. University of Fortaleza. October 2005

What Swarms can Teach Computer Scientists. University of Fortaleza, Brazil. April 2005

## STUDENT SUPERVISION

---

### CURRENT PHD STUDENTS

#### **Guilherme Machado: Causality in Human Mobility**

*University of Exeter*

2025 – Present

**Andrew Schendl: Using Mobility to Model Access to Greenspaces**  
*University of Exeter* 2023 – Present

**Julio Taveira: Using Network Science to Model Interstate Road Access in Brazil**  
*University of Exeter* 2023 – Present

**Guilherme Machado: Inequality in Social Media**  
*University of Exeter* 2022 – Present

#### SELECTED PAST PHD STUDENTS

**Liam Berrisford: Estimating Pollution from Secondary Public Datasets**  
*University of Exeter* 2020 – 2024

Currently a Research Software Engineer at the University of Exeter.

**Sima Farokhnejad: Modelling of Uncertainties in Cattle Epidemics**  
*University of Exeter* 2020 – 2024

Currently a Post-Doc at Southampton University, UK.

**Ana Jaramillo: Echo Chambers in Science**  
*University of Exeter* 2019 – 2023

Currently a post-doc at Technische Universität Graz, Austria.

**Clodomir Santana: Using Network Science to Characterise Swarm-based Algorithms**  
*University of Exeter* 2019 – 2023

Currently a post-doc at UC Davis, USA.

**Josemar F. Cruz: Social groups characterization and dynamics: a network science approach**  
*Florida Institute of Technology* 2014 – 2020

Currently a Data Scientist in Canada.

**Mahdi Seyednezhad: A Network Driven Approach for Characterizing Emoji Usage in Social Media**  
*Florida Institute of Technology* 2015 – 2020

Currently a Senior Data Scientist in the Chicago Region, USA.

**Diego P. Silva: A Comprehensive Data-Driven Characterization of Organ Transplantation**  
*Florida Institute of Technology* 2014 – 2018

Currently Assistant Professor at Universidade Católica de Pernambuco, Brazil.

**Diogo Pacheco: Information Densification of Social Constructs via Behavior Analysis of Social Media Users**  
*Florida Institute of Technology* 2013 – 2017

Currently a Senior Lecturer at the University of Exeter, UK.

**Basim Mahmood: Social Capital in Wireless Sensor Networks**  
*Florida Institute of Technology* 2012 – 2015

Currently an Associate Professor at the University of Mosul.

**Hugo Barbosa-Filho: Frequency and Recency of Visits in Human Mobility**

*Florida Institute of Technology*

2012 – 2016

Currently a Lecturer in Urban Systems, University of Exeter, UK.

**SERVICE ACTIVITIES**

---

**STEERING COMMITTEES AND BOARDS**

|                                                                          |                |
|--------------------------------------------------------------------------|----------------|
| Board Member, Latin American Conference on Complex Networks              | 2017 – Present |
| Co-Editor-in-Chief, Applied Network Science Journal, Springer Verlag     | 2015 – Present |
| Board Member, Complexity Series, Springer Verlag                         | 2016 – Present |
| Board Member, Complex Systems Society                                    | 2023 – Present |
| Associate Editor, Complexity Journal                                     | 2018 – Present |
| Editor, PLoS One                                                         | 2020 – Present |
| Associate Editor, Entropy Journal                                        | 2024 – Present |
| Co-Chair, Conference on Complex Networks (CompleNet)                     | 2009 – 2025    |
| Associate Editor, Learning & Nonlinear Models                            | 2016 – 2024    |
| Board Member, Network Science Society                                    | 2013 – 2025    |
| IEEE Task Force on Network Science                                       | 2011 – 2013    |
| ACM SIGAPP: Special Interest Group on Applied Computing                  | 2003 – 2009    |
| International Journal of Aviation Technology, Engineering and Management | 2011 – 2013    |

**RECENT CONFERENCE ORGANISATION**

|                                                                                          |                |
|------------------------------------------------------------------------------------------|----------------|
| General Co-Chair. CompleNet 2025. Fortaleza, Brazil.                                     | April 2025     |
| Program Co-Chair. Conference on Complex Systems. Exeter, UK.                             | September 2024 |
| General Co-Chair. CompleNet 2024. Exeter, UK.                                            | April 2024     |
| Organising Committee Member. NetSci-X. Venice, Italy.                                    | 2024           |
| Organising Committee Member. Conference on Complex Systems (CCS 2023). Salvador, Brazil. | October 2023   |
| Co-Chair. CompleNet 2023. Aveiro, Portugal.                                              | April 2023     |
| Co-Chair. GoodIT. Limassol, Cyprus.                                                      | 2022           |
| General Chair. CompleNet Live. Exeter, UK.                                               | 2020 – 2022    |
| General Chair. CompleNet. Exeter, UK.                                                    | 2020           |
| Program Chair. IDEAL. Manchester.                                                        | 2019           |
| Organising Committee Member. NetSci-X. Wroclaw, Poland.                                  | 2016           |
| Chair. NetSci-X. Rio de Janeiro.                                                         | 2015           |
| Chair. Workshop on Complex Networks (CompleNet).                                         | 2009 – Present |

REFERENCES

---

Available upon request.